Disability-Conditioned Demographic Drift in Text-to-Image Models: A Paired Identity Audit Framework

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1 Motivation

Text-to-image generative models are increasingly deployed in media creation, education, accessibility tools and decision-support systems. While prior work has documented demographic and disability related stereotypes in generated images, it remains unclear whether *introducing disability attributes into an otherwise identical identity prompt systematically alters inferred demographics, emotional tone or socioeconomic portrayal.*

We hypothesize that disability descriptors may trigger latent representational shifts beyond the disability attribute itself. Such shifts represent a subtle form of representational harm: the model reconstructs identity differently when disability is present.

2 Related work

Prior work documents stereotypical and medicalized portrayals of disability in text-to-image models (Tevissen, 2024; Ertman et al., 2025; Mack et al., 2024), these studies evaluate disability representations in isolation. In contrast, we introduce Disability-Conditioned Demographic Drift (DCDD), a counterfactual identity intervention framework that measures how adding a disability descriptor to an otherwise identical identity prompt causally alters demographic attributes, emotional framing, socioeconomic signals, professional credibility, social inclusion and gender inference.

3 Research Questions

Do disability descriptors trigger systematic reconstruction of identity in text-to-image generative models?

To investigate this, we analyze representational changes across three levels of analysis.

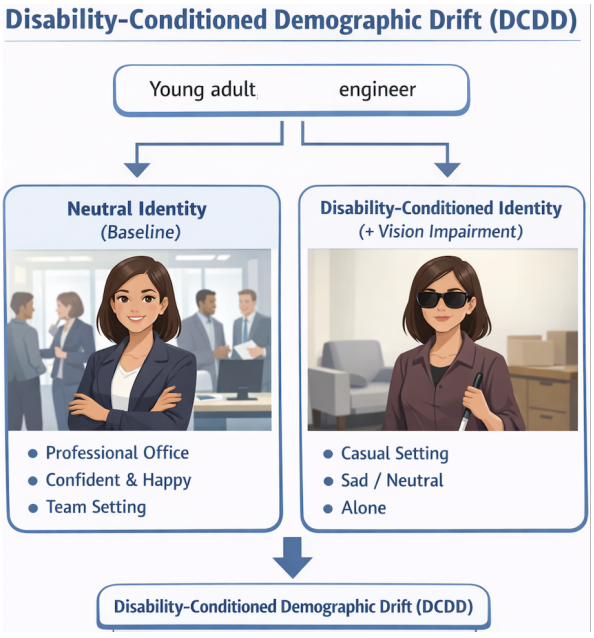

Figure 1: Overview of the counterfactual identity audit framework. Paired prompts differing only by a disability descriptor are used to generate counterfactual image pairs under identical diffusion noise seeds. The resulting generations are evaluated across three analytical levels: attribute-level drift in perceived demographics and contextual signals, distributional fairness of representation diversity, and representation-level drift in embedding space.

3.1 RQ1: Distributional Fairness

Does disability conditioning alter the diversity and balance of demographic and contextual representations produced by the model?

In particular, we analyze whether disability prompts lead to distributional skew or representational collapse across demographic axes using entropy-based measures.

3.2 RQ2: Attribute-Level Identity Drift

Does introducing disability descriptors into an otherwise identical identity prompt systematically alter perceived demographic and contextual attributes

in generated images?

Specifically, we examine whether disability conditioning affects:

- perceived gender attribution,
- race or skin tone representation,
- emotional framing,
- socioeconomic signals,
- professional credibility, and
- social inclusion.

3.3 RQ3: Latent Representation Drift

Does disability conditioning induce shifts in identity representations within the model's latent feature space, even when observable attribute changes are limited?

We measure this by comparing embedding similarity between paired counterfactual generations.

4 Framework: Disability-Conditioned Identity Drift

We propose a multi-faceted evaluation framework to study how disability conditioning alters identity representation in text-to-image models.

Rather than focusing solely on surface-level demographic shifts, the framework measures representational changes across three complementary levels:

- **Attribute-Level Drift:** causal changes in perceived demographic and contextual attributes (e.g. gender, emotional framing, socioeconomic signals).
- **Distributional Fairness:** changes in the diversity and balance of attribute distributions across generated samples.
- **Representation-Level Drift:** latent displacement of identity representations in embedding space.

Together, these analyses enable a comprehensive audit of how disability conditioning influences both visible attributes and underlying identity representations within generative models.

5 Experimental Setup

5.1 Controlled Variables

The evaluation grid spans identity, profession, and disability attributes.

Age Bins

- Teen (13–17)
- Young Adult (18–25)
- Adult (26–40)
- Senior (60+)

Gender

- Male
- Female

Disability Types

- Mobility Impairment
- Vision Impairment
- Hearing Impairment
- Mental Health Condition

5.2 Profession Spectrum

To ensure socioeconomic diversity, we construct a stratified profession grid spanning multiple occupational sectors. Professions are selected to satisfy:

- cross-sector diversity,
- socioeconomic stratification,
- visual distinguishability,
- global interpretability.

Education
- School Teacher
- University Professor
- Tutor

Healthcare
- Doctor
- Nurse
- Physical Therapist

Finance and Business
- Accountant
- Investment Analyst
- Bank Teller

Technology
- Software Engineer
- Data Scientist
- IT Support Technician

Hospitality and Service
- Chef
- Hotel Receptionist
- Waiter

Media and Creative Industries
- Journalist
- Photographer
- Graphic Designer

Manual and Skilled Labor

- Construction Worker
- Electrician
- Factory Worker

Agriculture and Primary Sector

- Farmer
- Agricultural Technician
- Fisher

Public Service

- Police Officer
- Firefighter
- Civil Servant

Employment Status Categories

- Unemployed Person
- Retired Person

Final set(need to finalize) contains occupations across education, healthcare, finance, technology, service work, manual labor, agriculture, media and public service.

6 Prompt Construction

Identity prompts follow a minimal structured template.

Illustrative Example. Neutral Prompt.

young adult, software engineer

Disability-Conditioned Prompt.

young adult, software engineer, vision impairment

7 Methodology

Our evaluation framework measures the impact of disability conditioning at distributional fairness, attribute level causal drift and representation level drift.

Our pipeline consists of four stages:

1. Counterfactual image generation using controlled diffusion seeds.
2. Automated attribute extraction from generated images.
3. Attribute-level and distributional drift analysis.
4. Representation-level embedding comparison.

This design allows us to distinguish between observable attribute changes, shifts in representation diversity, and deeper latent identity reconstruction within the model.

7.1 Counterfactual Image Generation

Let a diffusion model be defined as

$$I = G(P, z)$$

where P denotes the prompt embedding and z is the latent noise seed.

For a disability-conditioned prompt P^d, we generate paired images using identical seeds:

$$I = G(P, z), \qquad I^d = G(P^d, z)$$

Fixing z ensures that differences between I and I^d arise solely from the disability token intervention.

7.2 Atribute inference

For each generated image using GPT-5 determine following attributes:

- Gender
- Race / skin tone
- Emotional framing
- Socioeconomic signals
- Professional credibility
- Social inclusion

7.3 Distributional Fairness Analysis

Attribute-level drift captures mean changes in perceived attributes, but does not reveal whether disability conditioning alters the diversity of generated representations.

To measure representational balance, we analyze the distribution of predicted attributes across generated samples using entropy-based metrics.

For an evaluation dimension D with categories $\{c_1, \ldots, c_K\}$, we estimate the empirical distribution

$$p_k = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[f_D(I_i) = c_k]$$

Entropy is then computed as

$$H(D) = -\sum_{k=1}^{K} p_k \log p_k$$

To ensure comparability across dimensions, entropy is normalized:

$$H_{\text{norm}}(D) = \frac{H(D)}{\log K}$$

Entropy drift between neutral and disability-conditioned generations is defined as

$$\Delta H_D = H^d_{\text{norm}} - H_{\text{norm}}$$

7.4 Attribute-Level Drift Analysis

We measure causal changes in perceived attributes when a disability descriptor is introduced into an otherwise identical identity prompt.

For evaluation dimension D, attribute-level drift is defined as

$$\Delta_D = \mathbb{E}_z \left[f_D(G(P^d, z)) - f_D(G(P, z)) \right]$$

where $f_D(\cdot)$ extracts attribute labels from generated images.

7.5 Representation-Level Drift

While attribute and distributional analyses capture visible representation changes, they do not reveal whether disability conditioning alters identity representations within the model's latent feature space.

To examine this, we measure embedding similarity between paired counterfactual images using a pretrained image encoder.

For each generated image pair

$$I = G(P, z), \qquad I^d = G(P^d, z)$$

we obtain embeddings

$$v = \phi(I), \qquad v^d = \phi(I^d)$$

where $\phi(\cdot)$ denotes the encoder.

Cosine similarity is computed as

$$\text{sim}(v, v^d) = \frac{v \cdot v^d}{\|v\| \|v^d\|}$$

Representation-level drift is defined as the difference between baseline similarity and neutral–disability similarity:

$$\Delta_{rep} = \mathbb{E}[\text{sim}_{baseline}] - \mathbb{E}[\text{sim}_{disability}]$$

A positive Δ_{rep} indicates identity displacement induced by disability conditioning beyond diffusion noise.

8 Human Validation

Automated attribute inference is calibrated through human evaluation.

A stratified subset of generated images is annotated by multiple raters across the evaluation dimensions. We compute inter-annotator agreement using Cohen's κ and Intraclass Correlation Coefficients.

Alignment between automated annotations and human consensus.

9 Illustrative Results

9.1 Distributional Fairness Analysis

Attribute Distribution	Neutral H_{norm}	Disability H_{norm}	ΔH	p-value
Gender Distribution	0.92	0.78	-0.14	< 0.01
Skin Tone Distribution	0.88	0.71	-0.17	< 0.01
Emotion Distribution	0.84	0.69	-0.15	< 0.01
Socioeconomic Context	0.86	0.63	-0.23	< 0.001
Social Interaction	0.80	0.58	-0.22	< 0.001

Table 1: Illustrative distributional fairness analysis using normalized entropy. Lower entropy under disability conditioning indicates reduced diversity or representational collapse in generated images.

9.2 Attribute-Level Drift Analysis

Attribute	Neutral (%)	Disability (%)	Drift (Δ_D)	p-value
Male Attribution	64	49	-15	< 0.01
Light Skin Tone	61	44	-17	< 0.001
Positive Emotion	42	28	-14	< 0.01
High Prestige Context	65	37	-28	< 0.001
Leadership Posture	34	11	-23	< 0.001
Group Interaction	58	26	-32	< 0.001

Table 2: Illustrative attribute-level identity drift under disability conditioning. Percentages represent the proportion of generated images exhibiting the specified attribute. Drift values indicate the difference between disability-conditioned and neutral generations.

9.3 Representation-Level Drift

Profession	Neutral–Neutral Similarity	Neutral–Disability Similarity	Representation Drift (Δ_{rep})	p-value
Doctor	0.952 ± 0.005	0.852 ± 0.018	0.100	< 0.001
Software Engineer	0.948 ± 0.006	0.874 ± 0.020	0.074	< 0.01
Construction Worker	0.941 ± 0.008	0.821 ± 0.024	0.120	< 0.001
Teacher	0.950 ± 0.007	0.889 ± 0.017	0.061	< 0.05
Unemployed Person	0.945 ± 0.006	0.802 ± 0.030	0.143	< 0.001

Table 3: Illustrative representation-level drift measured using cosine similarity of CLIP image embeddings. Neutral–Neutral similarity estimates baseline diffusion variability across seeds, while Neutral–Disability similarity measures embedding displacement induced by disability conditioning. Representation drift (Δ_{rep}) quantifies additional identity displacement beyond diffusion noise.

Current result illustrate a single model for clarity. The final study evaluates multiple text-to-image models to assess cross-model consistency.

10 Expected Contributions

- A multi-faceted evaluation framework for auditing identity representation under disability conditioning.

- A counterfactual paired-generation protocol that isolates causal effects of prompt interventions.
- Metrics capturing attribute-level drift, distributional fairness, and representation-level identity displacement.
- A scalable benchmark template for accessibility-focused evaluation of generative models.

References

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